

Aayush Mishra

Software Engineer | C++ | Low-Latency Systems | Trading Infrastructure | Quantitative Technology | ☎ +91 8799980995 |

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Profile

Information Technology student with strong foundations in C++, data structures, algorithms, object-oriented programming, and Linux systems engineering. Experienced in designing and building high-performance, reliable, and reusable software — including exchange-style matching engines, low-latency order processing pipelines, risk management tooling, post-trade reconciliation workflows, and quantitative research platforms. Passionate about low-latency computing, high-throughput trading infrastructure, and scalable software architecture. Proven ability to manage multiple tasks in a fast-paced environment, communicate effectively across teams, and rapidly engineer production-quality solutions to challenging problems.

Education

St. John College of Engineering and Management ~ Maharashtra, India B.Tech in Information Technology | 2023 – 2027

- Coursework: Data Structures & Algorithms, Object-Oriented Programming, Operating Systems, Computer Networks, Database Management Systems, Concurrency & Multithreading, Artificial Intelligence, Machine Learning

Experience

Secretary – Training & Placement Cell ~ St. John College of Engineering and Management | Jun 2025 – Present

- Manage multiple concurrent workflows across 200+ students and 30+ recruiting companies in a fast-paced, deadline-driven environment — directly developing the task-management and communication skills essential in trading infrastructure teams
- Designed and built automated tracking pipelines and reporting dashboards, reducing manual overhead by 60% through reliable, reusable tooling
- Collaborate with cross-functional teams — faculty, recruiters, student leads — to assess operational needs and translate them into efficient technical workflows
- Drive execution of time-critical placement events, maintaining operational efficiency under pressure with strong communication and organizational discipline

AI & Machine Learning Intern ~ IBM | Sep 2025 – Oct 2025

- Built and optimized ML inference pipelines achieving 3× throughput improvement via batching, caching, and model quantization — techniques directly applicable to high-throughput market data processing systems
- Developed scalable, reusable software modules replacing brittle rule-based systems with ML-driven decision logic, reducing false positive rates by 40%
- Collaborated with cross-functional engineering and product teams to translate business requirements into technical implementations and ship production-ready software
- Gained hands-on exposure to performance benchmarking, system scalability, and software reliability in a production-adjacent environment

Machine Learning & AI Intern ~ Tamizhan Skills | Jun 2025 – Jul 2025

- Designed and implemented efficient, reusable ML pipelines for data ingestion, preprocessing, model training, and evaluation — applying strong software engineering discipline throughout
- Engineered modular feature extraction workflows in Python that reduced data preparation time by 70% versus prior manual processes
- Applied object-oriented design patterns, version control, and code review practices to deliver maintainable, scalable AI systems

Projects

Ultra Low-Latency Trading Engine Simulator | C++, Linux, Multithreading, Lock-Free Data Structures, OOP

- Designed, built, and maintained a high-performance, reliable trading engine in C++ — augmenting and re-implementing core components of an exchange-style low-latency/high-throughput order processing environment
- Built a fully reusable, efficient order book and matching engine supporting market, limit, cancel, and modify orders; implemented lock-free data structures using atomics and cache-aligned memory for <500ns median order processing latency
- Designed scalable multithreaded architecture with dedicated threads for market data ingestion, order matching, risk evaluation, and execution reporting — directly mirroring production trading infrastructure design
- Developed performance-tracking tools with latency histograms, throughput counters, and P99/P999 tail-latency monitoring for continuous regression detection and execution statistics analysis
- Applied memory pooling, object reuse, and zero-allocation critical paths — advanced C++ techniques essential for production low-latency/high-throughput trading environments
- Implemented risk-management modules performing real-time pre-trade limit checks, position exposure evaluation, and breach detection on every order event

AI-Augmented Risk Management & Post-Trade Reconciliation System | C++, Python, Machine Learning, SQL, Linux, Shell Scripts

- Designed and implemented scalable software solutions supporting trading operations, risk management, and post-trade life cycle workflows — including a state-of-the-art automated reconciliation system
- Built end-to-end post-trade reconciliation pipelines cross-referencing execution reports, trade confirmations, and settlement data; ML anomaly detection layer reduced unmatched trade investigation time by 65% versus traditional rule-based approaches
- Engineered real-time risk-management and performance-tracking tools monitoring position exposure, P&L attribution, drawdown breaches, and margin utilization — with ML-predicted breach alerts firing 2–3 minutes ahead of threshold hits
- Used C++ for latency-sensitive hot-path risk evaluation (limit checks, Greeks calculation) and Python with shell scripts for batch reporting, reconciliation automation, and operational workflows — a hybrid architecture mirroring production trading firm infrastructure
- Achieved 99.8% reconciliation match rate on synthetic trade datasets; demonstrated AI-augmented workflows are 4× faster at flagging discrepancies than purely manual or rule-based systems
- Collaborated across simulated cross-functional requirements — trading, risk, operations — to translate business needs into reliable, reusable, and maintainable technical implementations

Quantitative Research & Backtesting Platform with AI-Powered Market Microstructure Analyzer | Python, C++, SQL, Linux, Machine Learning, Shell Scripts

- Designed and implemented a scalable quantitative research platform supporting factor analysis, strategy development, and portfolio-level backtesting — enabling faster, more reliable trading strategy iteration than traditional manual approaches
- Built high-performance C++ data ingestion layer processing 500K+ simulated market events per second, feeding Python-based ML inference and analytics pipelines — efficient, reusable architecture spanning the full market analysis workflow
- Developed ML models (gradient boosting + LSTM) analyzing order-book dynamics and short-term market behavior from Level 2 data; engineered 40+ microstructure features including queue depletion rates, spread dynamics, and volume-weighted order flow imbalance
- Achieved 58% directional accuracy on 10-second price moves ($p < 0.01$, 6-month walk-forward validation); AI-driven approach reduced signal discovery time by ~80% versus traditional chart-based analysis
- Implemented momentum, mean-reversion, and statistical arbitrage strategies with full transaction cost modeling, slippage simulation, and automated performance-tracking tools (Sharpe, Sortino, max drawdown, Calmar ratio, factor exposure)
- Integrated ML-based regime detection (Hidden Markov Models) for dynamic parameter adjustment; backtested 50+ strategy variants and identified 3 robust signals (Sharpe > 1.5) confirmed by walk-forward validation

Technical Skills

Languages: C++ (Primary), Python, Java, SQL, OCaml, Bash / Shell Scripting

Core CS: Data Structures & Algorithms, Object-Oriented Programming, Design Patterns, Concurrency & Multithreading, Lock-Free Programming, Memory Management, Reusable & Reliable Code Design

Systems & Infrastructure: Linux (primary development environment), Git, GitHub, Docker, REST APIs, POSIX Threads, Ring Buffers, WebSockets, IPC, Shell Scripts

Trading & Quantitative: Low-Latency Systems, High-Throughput Pipelines, Order Book Mechanics, Matching Engine Design, Market Microstructure, Risk Management, Performance Tracking, Post-Trade Reconciliation, Backtesting, Statistical Arbitrage, Factor Research

AI & Machine Learning: PyTorch, scikit-learn, OpenCV, Model Quantization, TorchScript, HMM, Gradient Boosting, LSTM, Feature Engineering, ML Pipeline Design

Achievements

- Secretary, Training & Placement Cell — managing placement operations for 200+ students across 30+ companies in a fast-paced environment
- Built 3+ production-quality systems directly targeting low-latency trading infrastructure, risk management, post-trade reconciliation, and quantitative research
- Demonstrated AI-augmented engineering workflows deliver 4–80× speed improvements over traditional approaches across reconciliation, signal discovery, and market analysis
- Research contributor: AI-driven safety intelligence framework for large-scale infrastructure (RailSafe)
- Active participant in technical competitions, engineering hackathons, and systems programming challenges

Languages

English (Fluent), Hindi (Native), Marathi (Native), Japanese (Beginner), German (Beginner)